Akoizo Original Practice - Form 2 - Answer key

Division B | Boomilever | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) In a typical Boomilever, which member is in tension?

Unit: Statics of a cantilever: moments, tension, and compression.
Worked-case data: A 15 kg load hangs 42 cm from the wall. The strut meets the wall 20 cm below the bolt. Estimate the horizontal chord force.
Separate investigation observations:
Test A: Strut 15 cm deep, 3 mm square: bowed sideways at 9 kg.
Test B: Same strut with one mid-length brace: held 15 kg.
Test C: Chord 1.5 x 3 mm: never failed in any test.
Test D: Chord-to-bolt joint glued with a single drop: pulled off at 12 kg.

A. The Loading Block

B. Every member equally

C. The top chord that hooks on the bolt

D. The strut pressing on the wall

Correct: C

2. (2 points) A strut bowed sideways and then snapped in the middle. What is the best next change?

A. Add a brace at mid-length

B. Use thicker chord wood

C. Add glue to the hook

D. Increase the reach

Correct: A

3. (2 points) Before sand is added, the bottom of the Loading Block must be:

Unit: Joints, adhesives, and the bolt hook.
Worked-case data: A chord joint must carry 300 N. Your glue holds about 2 N per mm^2 of long-grain overlap. How much bonded area do you need, with a safety factor of 1.5?
Separate investigation observations:
Failure 1: Hook crushed into an oval around the bolt at 10 kg.
Failure 2: Strut-to-chord butt joint split at 13 kg.
Fix trial: Basswood bearing pad (+0.15 g) and two gussets (+0.10 g): held 15 kg.

A. Above the horizontal plane through the bolt centerline

B. Below the bolt plane

C. Touching the wall

D. Exactly 45 cm from the wall

Correct: A

4. (4 points) Using T = W x L ÷ d, find the chord force for 147 N at L = 0.40 m and d = 0.20 m.

answer: T = 147 x 0.40 ÷ 0.20 about 294 N. (4 points).

5. (4 points) Why might a team choose the base option even though the bonus is large?

answer: A shallower wall contact makes member forces larger, so the bonus structure may need much more wood, and it risks failing before 15 kg and losing the bonus anyway. (4 points).

6. (4 points) Why should a team build a practice wall matching the official dimensions?

answer: The hook, contact foot, and reach depend on exact bolt clearance, bolt height, and contact lines, so practice on matching geometry reveals fit and rule problems before competition. (4 points).

7. (4 points) Explain why a box or spaced-pair strut beats a solid square of the same mass.

answer: Moving material away from the center raises the second moment of area I much faster than it raises area, so buckling resistance rises for the same mass. (4 points).

8. (4 points) Why does a flatter strut need to push harder?

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answer: Only the vertical component of the strut force supports the load; a flatter angle gives a smaller vertical share, so the total force must be larger. (4 points).

9. (4 points) How can slow-motion video help failure analysis?

Unit: Testing, failure analysis, and iteration.
Worked-case data: Your log shows: 9.2 g held 15 kg; 8.4 g held 15 kg; 7.9 g failed at 13.1 kg (strut buckled). What should the competition build be?
Separate investigation observations:
Builds 1-3: 8.1 g, 8.3 g, 8.2 g: all held 15 kg.
Builds 4-5: 7.4 g, 7.5 g: failed at 12 kg and 14 kg, strut buckled.
Build 6: 7.6 g with a stiffer, denser strut: held 15 kg.

answer: It shows which part failed first, since secondary breaks happen as the structure collapses and can mislead you. (4 points).

10. (4 points) Why must the hook fit within the bolt's 3.0 cm clearance?

answer: The bolt sticks out only about 3.0 cm from the wall, so the hook and any bearing surface must sit on that exposed shaft without touching the nut or washer region improperly. (4 points).

11. (2 points) Define "Reach" in the context of Boomilever.

answer: Horizontal distance from wall to chain centerline: 40-45 cm. (2 points).

12. (2 points) Define "Tier 2" in the context of Boomilever.

answer: Placement for structures that break one or more rules; ranks below all of Tier 1. (2 points).

13. (2 points) What most likely caused the difference?

Supplied synthetic observations:
Run 1: Setup took 5 min; fast pour; failed at 9 kg with violent swing.
Run 2: Setup took 2 min; slow pour with stick tips; held 15 kg.
Inspection: Both structures were identical builds from the same wood batch.

A. Different wood in Run 2

B. The bonus line

C. Rushed loading and bucket swing in Run 1

Correct: C

14. (2 points) Define "Density" in the context of Boomilever.

answer: Mass per volume, g/cm^3; strongly linked to balsa stiffness. (2 points).

15. (2 points) Define "Compression member" in the context of Boomilever.

answer: A member being pushed; usually the lower strut, at risk of buckling. (2 points).

16. (2 points) Define "Controlled variable" in the context of Boomilever.

answer: A factor kept the same so one change can be judged. (2 points).

17. (2 points) Define “Estimated Load Supported” in the context of Boomilever.

answer: The team’s pre-test prediction, used as a tiebreaker. (2 points).

18. (2 points) Define “End grain” in the context of Boomilever.

answer: The cut end of wood fibers; bonds poorly and absorbs glue. (2 points).